3 Canonical Forms

3.1 Jordan Forms & Generalized Eigenvectors

Essentially everything in these notes is a variation of a single question: given a linear map T on a finite-dimensional vector space V, how can we find a basis β such that the matrix $[\mathrm{T}]_\beta$ is as close to diagonal as possible? In this chapter we see what is possible when T is non-diagonalizable.

Example 3.1. The matrix $A = \left(\begin{smallmatrix} -8 & 4 \\ -25 & 12 \end{smallmatrix}\right) \in M_2(\mathbb{R})$ has characteristic equation

$$p(t) = (-8-t)(12-t) + 4 \cdot 25 = t^2 - 4t + 4 = (t-2)^2$$

and thus a single eigenvalue $\lambda = 2$ of algebraic multiplicity two. The matrix is non-diagonalizable since the corresponding eigenspace is one-dimensional (Theorem 1.6):

$$E_2 = \mathcal{N}\begin{pmatrix} -10 & 4 \\ -25 & 10 \end{pmatrix} = \operatorname{Span}\begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

However, with respect to a basis $\beta = \{\mathbf{v}_1, \mathbf{v}_2\}$ where $\mathbf{v}_1 = \left(\begin{smallmatrix} 2 \\ 5 \end{smallmatrix}\right)$ is an eigenvector, $[\mathrm{L}_A]_\beta$ is *upper-triangular* which is better than nothing! How simple can we make this matrix? If $\mathbf{v}_2 = \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$, then

$$\begin{aligned} A\mathbf{v}_2 &= \begin{pmatrix} -8x + 4y \\ -25x + 12y \end{pmatrix} = 2\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -10x + 4y \\ -25x + 10y \end{pmatrix} = 2\mathbf{v}_2 + (-5x + 2y)\mathbf{v}_1 \\ \implies [\mathrm{L}_A]_\beta &= \begin{pmatrix} 2 & -5x + 2y \\ 0 & 2 \end{pmatrix} \end{aligned}$$

Since $\mathbf{v}_2$ cannot be parallel to $\mathbf{v}_1$, the only thing we cannot have is a diagonal matrix. The next best thing is to make the upper right corner 1; this can be achieved by, for instance, choosing

$$\beta = \{\mathbf{v}_1, \mathbf{v}_2\} = \left\{\begin{pmatrix} 2 \\ 5 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \end{pmatrix}\right\} \implies [\mathrm{L}_A]_\beta = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

Definition 3.2. A *Jordan block* is a square matrix of the form

$$J = \begin{pmatrix} \lambda & 1 & & \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{pmatrix}$$

where all non-indicated entries are zero. Any 1×1 matrix is also a Jordan block.
A *Jordan canonical form* is a block-diagonal matrix $\operatorname{diag}(J_1, \ldots, J_m)$ where each J_k is a Jordan block.
A *Jordan canonical basis* for $\mathrm{T} \in \mathcal{L}(V)$ is a basis β of V such that $[\mathrm{T}]_\beta$ is a Jordan canonical form.[17]

In Example 3.1, β is a Jordan canonical basis and $[\mathrm{L}_A]_\beta$ a Jordan canonical form.

If a map is diagonalizable, then any eigenbasis is Jordan canonical and the corresponding Jordan canonical form is diagonal. What about more generally? Does every non-diagonalizable map have a Jordan canonical basis? If so, how can we find such?

[17] If $\mathrm{T} = \mathrm{L}_A$ is left-multiplication by a matrix $A \in M_n(\mathbb{F})$, we simply refer to a Jordan canonical basis for A.

Example 3.3. We check that $\beta = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \left\{ \left(\begin{smallmatrix} 1 \\ 0 \\ 1 \end{smallmatrix}\right), \left(\begin{smallmatrix} 1 \\ 2 \\ 0 \end{smallmatrix}\right), \left(\begin{smallmatrix} 1 \\ 1 \\ 1 \end{smallmatrix}\right) \right\}$ is a Jordan canonical basis for

$$A = \begin{pmatrix} -1 & 2 & 3 \\ -4 & 5 & 4 \\ -2 & 1 & 4 \end{pmatrix}$$

Indeed

$$A\mathbf{v}_1 = 2\mathbf{v}_1, \quad A\mathbf{v}_2 = 3\mathbf{v}_2, \quad A\mathbf{v}_3 = \begin{pmatrix} 4 \\ 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 1+3 \\ 2+3 \\ 0+3 \end{pmatrix} = \mathbf{v}_2 + 3\mathbf{v}_3 \implies [\mathrm{L}_A]_\beta = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{pmatrix}$$

This last is a Jordan canonical form for A.

Generalized Eigenvectors

Example 3.3 was easy to verify, but how would we go about finding a suitable basis in general? We brute-forced this in Example 3.1, but this is not a reasonable way to proceed for harder problems. Considering the shape of a Jordan form, any basis vector corresponding to a column with at most one non-zero entry must be an eigenvector:

- In Example 3.1, $\mathbf{v}_1$ is an eigenvector but $\mathbf{v}_2$ is not.
- In Example 3.3, $\mathbf{v}_1$ and $\mathbf{v}_2$ are eigenvectors but $\mathbf{v}_3$ is not.

The practical question, therefore, is how to fill out a Jordan canonical basis once we have a maximal independent set of eigenvectors. We now define the necessary objects.

Definition 3.4. Suppose $\mathrm{T} \in \mathcal{L}(V)$ has an eigenvalue λ. Its *generalized eigenspace* is

$$K_\lambda := \{\mathbf{x} \in V : (\mathrm{T} - \lambda\mathrm{I})^k(\mathbf{x}) = \mathbf{0} \text{ for some } k \in \mathbb{N}\} = \bigcup_{k \in \mathbb{N}} \mathcal{N}(\mathrm{T} - \lambda\mathrm{I})^k$$

A *generalized eigenvector* is any non-zero $\mathbf{v} \in K_\lambda$.

As with eigenspaces, the generalized eigenspaces of $A \in M_n(\mathbb{F})$ are those of the map $\mathrm{L}_A \in \mathcal{L}(\mathbb{F}^n)$.

Examples (cont.). Our earlier Jordan canonical bases consist of generalized eigenvectors.

Example 3.1. We have one eigenvalue $\lambda = 2$. Since $(A - 2I)^2 = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ is the zero matrix, every non-zero vector is a generalized eigenvector: that is, $K_2 = \mathbb{R}^2$.

Example 3.3. We see that

$$(A - 2I)\mathbf{v}_1 = \mathbf{0}, \qquad (A - 3I)\mathbf{v}_2 = \mathbf{0}, \qquad (A - 3I)^2\mathbf{v}_3 = (A - 3I)\mathbf{v}_2 = \mathbf{0}$$

whence β is a basis of generalized eigenvectors. Indeed

$$K_3 = \mathrm{Span}\{\mathbf{v}_1, \mathbf{v}_2\}, \quad K_2 = E_2 = \mathrm{Span}\{\mathbf{v}_3\}$$

though verifying this with current technology is a little awkward…

To help compute generalized eigenspaces, we invoke the main result of this section; the proof is postponed due to its meatiness. Make sure to compare this with the diagonalizability criterion (Theorem 1.13).

Theorem 3.5. *Suppose that the characteristic polynomial of* $\mathrm{T} \in \mathcal{L}(V)$ *splits over* $\mathbb{F}$:

$$p(t) = (\lambda_1 - t)^{m_1} \cdots (\lambda_k - t)^{m_k}$$

where the m_j *are the algebraic multiplicities of the* distinct *eigenvalues* λ_j. *Then:*

1. *For each eigenvalue, (a)* $K_\lambda = \mathcal{N}(\mathrm{T} - \lambda \mathrm{I})^m$, *and (b)* $\dim K_\lambda = m$.
2. $V = K_{\lambda_1} \oplus \cdots \oplus K_{\lambda_k}$. *In particular, there exists a basis of generalized eigenvectors.*

With regard to part 2, we shall eventually be able to *choose* the basis to be Jordan canonical. In conclusion: a map has a Jordan canonical basis if and only if its characteristic polynomial splits.

Examples 3.6. We revisit the previous example in this new language, and consider a second matrix from scratch.

1. (Example 3.3) $A = \left(\begin{smallmatrix} -1 & 2 & 3 \\ -4 & 5 & 4 \\ -2 & 1 & 4 \end{smallmatrix}\right)$ has characteristic polynomial $p(t) = (2-t)^1(3-t)^2$. Therefore,

$$K_2 = \mathcal{N}(A - 2I)^1 = \operatorname{Span}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \implies \dim K_2 = 1$$

$$K_3 = \mathcal{N}(A - 3I)^2 = \mathcal{N}\begin{pmatrix} 2 & -1 & -1 \\ 0 & 0 & 0 \\ 2 & -1 & -1 \end{pmatrix} = \operatorname{Span}\left\{\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right\} \implies \dim K_3 = 2$$

   In accordance with the theorem, $\mathbb{R}^3 = K_2 \oplus K_3$.

2. We find the generalized eigenspaces of the matrix $B = \left(\begin{smallmatrix} 5 & 2 & -1 \\ 0 & 0 & 0 \\ 9 & 6 & -1 \end{smallmatrix}\right)$.

   The characteristic polynomial is

$$p(t) = \det(B - \lambda I) = -t\begin{vmatrix} 5-t & -1 \\ 9 & -1-t \end{vmatrix} = -t(t^2 - 5t + t - 5 + 9) = -(0-t)^1(2-t)^2$$

   $\operatorname{mult} 0 = 1$: Indeed $K_0 = \mathcal{N}(B - 0I)^1 = \mathcal{N}(B) = \operatorname{Span}\left(\begin{smallmatrix} 1 \\ -1 \\ 3 \end{smallmatrix}\right)$ is the eigenspace E_0.

   $\operatorname{mult} 2 = 2$: This time, the generalized eigenspace is

$$K_2 = \mathcal{N}(B - 2I)^2 = \mathcal{N}\begin{pmatrix} 3 & 2 & -1 \\ 0 & -2 & 0 \\ 9 & 6 & -3 \end{pmatrix}^2 = \mathcal{N}\begin{pmatrix} 0 & -4 & 0 \\ 0 & 4 & 0 \\ 0 & -12 & 0 \end{pmatrix} = \operatorname{Span}\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right\}$$

   The corresponding eigenspace $E_2 = \operatorname{Span}\left(\begin{smallmatrix} 1 \\ 0 \\ 3 \end{smallmatrix}\right)$ is a *one*-dimensional subspace of K_2. The matrix is non-diagonalizable.

   Observe also that $\mathbb{R}^3 = K_0 \oplus K_2$.

Properties of Generalized Eigenspaces and the Proof of Theorem 3.5

A lot of work is required to justify the main result. Feel free to skip the proofs at first reading.

Lemma 3.7. *Let λ be an eigenvalue of $\mathrm{T} \in \mathcal{L}(V)$. Then:*

1. *E_λ is a subspace of K_λ, which is itself a subspace of V.*
2. *K_λ is T-invariant.*
3. *Suppose K_λ is finite-dimensional and $\mu \neq \lambda$. Then:*
   (a) *K_λ is $(\mathrm{T} - \mu\mathrm{I})$-invariant and the restriction of $\mathrm{T} - \mu\mathrm{I}$ to K_λ is an isomorphism.*
   (b) *If μ is another eigenvalue, then $K_\lambda \cap K_\mu = \{\mathbf{0}\}$. In particular K_λ contains no eigenvectors other than those in E_λ.*

Proof. Some of the details are left to Exercise 6.

1. These are an exercise.
2. Let $\mathbf{x} \in K_\lambda$. We must show that $\mathrm{T}(\mathbf{x}) \in K_\lambda$. We know $\exists k$ such that $(\mathrm{T} - \lambda\mathrm{I})^k(\mathbf{x}) = \mathbf{0}$, but then

$$\begin{aligned}(\mathrm{T} - \lambda\mathrm{I})^k\big(\mathrm{T}(\mathbf{x})\big) &= (\mathrm{T} - \lambda\mathrm{I})^k\big(\mathrm{T}(\mathbf{x}) - \lambda\mathbf{x} + \lambda\mathbf{x}\big) \\ &= (\mathrm{T} - \lambda\mathrm{I})^{k+1}(\mathbf{x}) + \lambda(\mathrm{T} - \lambda\mathrm{I})^k(\mathbf{x}) = \mathbf{0}\end{aligned}$$

   Otherwise said, $\mathrm{T}(\mathbf{x}) \in K_\lambda$.

3. (a) Since K_λ is T-invariant (part 2), it is also $(\mathrm{T} - \mu\mathrm{I})$-invariant: if $\mathbf{x} \in K_\lambda$, then

$$(\mathrm{T} - \mu\mathrm{I})(\mathbf{x}) = \mathrm{T}(\mathbf{x}) - \mu\mathbf{x} \in K_\lambda$$

   Suppose, for contradiction, that $\mathrm{T} - \mu\mathrm{I}$ is *not injective* on K_λ. Then

$$\exists \mathbf{y} \in K_\lambda \setminus \{\mathbf{0}\} \text{ such that } (\mathrm{T} - \mu\mathrm{I})(\mathbf{y}) = \mathbf{0}$$

   Let $k \in \mathbb{N}$ be **minimal** such that $(\mathrm{T}-\lambda\mathrm{I})^k(\mathbf{y}) = \mathbf{0}$ ($\mathbf{y} \in K_\lambda$!) and define $\mathbf{z} = (\mathrm{T}-\lambda\mathrm{I})^{k-1}(\mathbf{y}) \neq \mathbf{0}$. We compute both $(\mathrm{T} - \lambda\mathrm{I})(\mathbf{z})$ and $(\mathrm{T} - \mu\mathrm{I})(\mathbf{z})$:
   - $(\mathrm{T} - \lambda\mathrm{I})(\mathbf{z}) = (\mathrm{T} - \lambda\mathrm{I})^k(\mathbf{y}) = \mathbf{0}$ says $\mathbf{z}$ is an eigenvector with eigenvalue λ.
   - Since $\mathrm{T} - \mu\mathrm{I}$ and $\mathrm{T} - \lambda\mathrm{I}$ commute (easy exercise),

$$\begin{aligned}(\mathrm{T} - \mu\mathrm{I})(\mathbf{z}) &= (\mathrm{T} - \mu\mathrm{I})(\mathrm{T} - \lambda\mathrm{I})^{k-1}(\mathbf{y}) \\ &= (\mathrm{T} - \lambda\mathrm{I})^{k-1}(\mathrm{T} - \mu\mathrm{I})(\mathbf{x}) = \mathbf{0}\end{aligned}$$

     whence $\mathbf{z}$ is also an eigenvector with eigenvalue μ! If μ isn't an eigenvalue, then we already have a contradiction. Even if it is, $E_\mu \cap E_\lambda = \{\mathbf{0}\}$ contradicts $\mathbf{z} \neq \mathbf{0}$.

   We conclude that the restriction $(\mathrm{T} - \mu\mathrm{I})_{K_\lambda} \in \mathcal{L}(K_\lambda)$ is injective. Since $\dim K_\lambda < \infty$, the restriction is automatically an isomorphism.

   (b) This is another exercise.

■

Proof of Theorem 3.5. Assume that the characteristic polynomial splits: $p(t) = (\lambda_1 - t)^{m_1} \cdots (\lambda_k - t)^{m_k}$.

1. (a) Fix an eigenvalue λ. By definition, $\mathcal{N}(\mathrm{T} - \lambda\mathrm{I})^m \le K_\lambda = \bigcup_{n\in\mathbb{N}} \mathcal{N}(\mathrm{T} - \lambda\mathrm{I})^n$.

   The opposite inclusion follows from Lemma 3.7). Since K_λ contains no eigenvectors from any eigenvalue other than λ, the characteristic polynomial of the restriction T_{K_λ} must be

   $$p_{K_\lambda}(t) = (\lambda - t)^{\dim K_\lambda}$$

   By T-invariance, $p_{K_\lambda}(t)$ divides $p(t)$ (Theorem 1.11), whence $\dim K_\lambda \le m$. By Cayley–Hamilton (Theorem 1.20), $p_{K_\lambda}(\mathrm{T}_{K_\lambda}) = 0$, whence

   $$\forall \mathbf{x} \in K_\lambda,\ (\lambda\mathrm{I} - \mathrm{T})^{\dim K_\lambda}(\mathbf{x}) = \mathbf{0} \implies K_\lambda \le \mathcal{N}(\mathrm{T} - \lambda\mathrm{I})^m$$

1. (b) and 2. We prove simultaneously by induction[18] on the number of distinct eigenvalues of T.

   Base case If T has only one eigenvalue, then $p(t) = (\lambda - t)^m$ where $m = \dim V$. Cayley–Hamilton says that $(\mathrm{T} - \lambda\mathrm{I})^m(\mathbf{x}) = \mathbf{0}$ for all $\mathbf{x} \in V$. Thus $V = K_\lambda$ and $\dim K_\lambda = m$.

   Induction step Fix k and suppose the results hold for all maps with k distinct eigenvalues. Let T have distinct eigenvalues $\lambda_1, \ldots, \lambda_k, \mu$, with multiplicities $m_1, \ldots, m_k, m$ respectively. Define

   $$W := \mathcal{R}(\mathrm{T} - \mu\mathrm{I})^m$$

   The subspace W has several properties, the first two of which are left to Exercise 7:

   - W is T-invariant.
   - $W \cap K_\mu = \{\mathbf{0}\}$ so that μ is not an eigenvalue of the restriction T_W.
   - Each $K_{\lambda_j} \le W$: since $(\mathrm{T} - \mu\mathrm{I})_{K_{\lambda_j}}$ is an isomorphism (Lemma part 3), we can invert,

   $$\mathbf{x} \in K_{\lambda_j} \implies \mathbf{x} = (\mathrm{T} - \mu\mathrm{I})^m \Big((\mathrm{T} - \mu\mathrm{I})_{K_{\lambda_j}}^{-1} \Big)^m (\mathbf{x}) \in \mathcal{R}(\mathrm{T} - \mu\mathrm{I})^m = W$$

   We conclude that each λ_j is an eigenvalue of the restriction T_W with generalized eigenspace K_{λ_j}. Since T_W has k distinct eigenvalues, the induction hypotheses apply:

   $$W = K_{\lambda_1} \oplus \cdots \oplus K_{\lambda_k} \quad \text{and} \quad p_W(t) = (\lambda_1 - t)^{\dim K_{\lambda_1}} \cdots (\lambda_k - t)^{\dim K_{\lambda_k}}$$

   Since $W \cap K_\mu = \{\mathbf{0}\}$, we finish by using the rank–nullity theorem to count dimensions:

   $$\begin{aligned} \dim V &= \operatorname{rank}(\mathrm{T} - \mu\mathrm{I})^m + \operatorname{null}(\mathrm{T} - \mu\mathrm{I})^m = \dim W + \dim K_\mu = \sum_{j=1}^k \dim K_{\lambda_j} + \dim K_\mu \\ &\le m_1 + \cdots + m_k + m = \deg(p(t)) = \dim V \end{aligned}$$

   We therefore have *equality*; each $\dim K_{\lambda_j} = m_j$ and $\dim K_\mu = m$. We conclude that

   $$V = K_{\lambda_1} \oplus \cdots \oplus K_{\lambda_k} \oplus K_\mu$$

   which completes the induction step and thus the proof. Whew! ■

[18] This is yet another argument where we consider a suitable subspace to which we can apply an induction hypothesis; recall the spectral theorem, Schur's lemma, bilinear form diagonalization, etc. Theorem 3.12 will provide one more!

Cycles of Generalized Eigenvectors

Theorem 3.5 says that every linear map whose characteristic polynomial splits has a generalized eigenbasis. This isn't quite the same as a Jordan canonical basis, but we're very close!

Example 3.8. The matrix $A = \begin{pmatrix} 5&1&0\\0&5&1\\0&0&5 \end{pmatrix} \in M_3(\mathbb{R})$ is a single Jordan block. There is a single generalized eigenspace $K_5 = \mathbb{R}^3$, and the standard basis $\epsilon = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is Jordan canonical.
The crucial observation for what follows is that one of these vectors $\mathbf{e}_3$ *generates* the others via repeated applications of $A - 5I$:

$$\mathbf{e}_2 = (A-5I)\mathbf{e}_3, \qquad \mathbf{e}_1 = (A-5I)\mathbf{e}_2 = (A-5I)^2\mathbf{e}_3$$

In the arguments and examples that follow, it will be helpful to refer to the linear map $\mathrm{U} = \mathrm{T} - \lambda\mathrm{I}$, where λ is an eigenvalue of T. Note the following:

- The nullspace of U is the *eigenspace*: $\mathcal{N}(\mathrm{U}) = E_\lambda \leq K_\lambda$.
- T and U commute, that is $\mathrm{TU} = \mathrm{UT}$.

Definition 3.9. Let T have eigenvalue λ, suppose $\mathbf{x} \in K_\lambda$ is a generalized eigenvector and let k be minimal such that $(\mathrm{T}-\lambda\mathrm{I})^k(\mathbf{x}) = \mathbf{0}$. The *cycle of generalized eigenvectors generated by* $\mathbf{x}$ is the set

$$\beta_{\mathbf{x}} := \left\{(\mathrm{T}-\lambda\mathrm{I})^{k-1}(\mathbf{x}), \ldots, (\mathrm{T}-\lambda\mathrm{I})(\mathbf{x}), \mathbf{x}\right\} = \left\{\mathrm{U}^{k-1}(\mathbf{x}), \ldots, \mathrm{U}(\mathbf{x}), \mathbf{x}\right\}$$

Note that only the first element $(\mathrm{T}-\lambda\mathrm{I})^{k-1}(\mathbf{x})$ of $\beta_{\mathbf{x}}$ is an *eigenvector*. Moreover, $\operatorname{Span}\beta_{\mathbf{x}} = \langle \mathbf{x} \rangle$ is the U-cyclic subspace generated by $\mathbf{x}$.
Our goal is to show that K_λ has a basis consisting of cycles of generalized eigenvectors; putting these together results in a Jordan canonical basis.

Lemma 3.10. *Let $\beta_{\mathbf{x}}$ be a cycle of generalized eigenvectors of* T *with length k. Then:*

1. *$\beta_{\mathbf{x}}$ is linearly independent and thus a basis of* $\operatorname{Span}\beta_{\mathbf{x}}$ *(which necessarily has dimension k).*
2. $\operatorname{Span}\beta_{\mathbf{x}}$ *is* T*-invariant. With respect to $\beta_{\mathbf{x}}$, the matrix of the restriction of* T *is the $k \times k$ Jordan block* $[\mathrm{T}_{\operatorname{Span}\beta_{\mathbf{x}}}]_{\beta_{\mathbf{x}}} = \begin{pmatrix} \lambda & 1 & & \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{pmatrix}$.

Proof. As indicated above, we write $\mathrm{U} = \mathrm{T} - \lambda\mathrm{I}$.

1. Feed the linear combination $\sum_{j=0}^{k-1} a_j \mathrm{U}^j(\mathbf{x}) = \mathbf{0}$ to U^{k-1} to obtain

$$a_0\mathrm{U}^{k-1}(\mathbf{x}) = \mathbf{0} \implies a_0 = 0 \qquad \text{(remember } \mathrm{U}^k(\mathbf{x}) = \mathbf{0}!\text{)}$$

   Now feed the same linear combination to U^{k-2}, etc., to see that all coefficients $a_j = 0$.

2. Since T and U commute,

$$\mathrm{T}\big(\mathrm{U}^j(\mathbf{x})\big) = \mathrm{U}^j\big(\mathrm{T}(\mathbf{x})\big) = \mathrm{U}^j\big((\mathrm{U}+\lambda\mathrm{I})(\mathbf{x})\big) = \mathrm{U}^{j+1}(\mathbf{x}) + \lambda\mathrm{U}^j(\mathbf{x}) \in \operatorname{Span}\beta_{\mathbf{x}}$$

   This justifies both T-invariance and the Jordan block claim!

∎

Once we have the generalized eigenspaces of T, the naïve approach to finding a Jordan canonical basis is to play with cycles until we find a basis for each K_λ. *Many* choices of canonical basis exist for a given map! We'll consider a more systematic method in the next section.

Examples 3.11. We compute Jordan canonical bases for three linear maps/matrices.

1. The characteristic polynomial $p(t) = (1-t)^3$ of $A = \left(\begin{smallmatrix} 1 & 0 & 2 \\ 0 & 1 & 6 \\ 6 & -2 & 1 \end{smallmatrix}\right)$ splits over $\mathbb{R}$.

   We hack our way to a Jordan canonical basis by choosing some $\mathbf{x} \in K_1 = \mathbb{R}^3$ and considering what $U = A - I$ does to it! For instance, with $\mathbf{x} = \mathbf{e}_1$,

   $$\beta_{\mathbf{x}} = \left\{ \mathrm{U}^2 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \mathrm{U} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\} = \left\{ \begin{pmatrix} 12 \\ 36 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}$$

   is a Jordan canonical basis. Otherwise said,

   $$A = QJQ^{-1} = \begin{pmatrix} 12 & 0 & 1 \\ 36 & 0 & 0 \\ 0 & 6 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 12 & 0 & 1 \\ 36 & 0 & 0 \\ 0 & 6 & 0 \end{pmatrix}^{-1}$$

   In practice, almost any choice of $\mathbf{x} \in \mathbb{R}^3$ will generate a cycle of length three and thus a basis!

2. The matrix $B = \left(\begin{smallmatrix} 7 & 1 & -4 \\ 0 & 3 & 0 \\ 8 & 1 & -5 \end{smallmatrix}\right) \in M_3(\mathbb{R})$ has characteristic polynomial $p(t) = -(t+1)^{1}(t-3)^{2}$.

   Since $\dim K_{-1} = 1$, we see that $K_{-1} = E_{-1} = \operatorname{Span}\left(\begin{smallmatrix} 1 \\ 0 \\ 2 \end{smallmatrix}\right)$ is spanned by a cycle of length one.

   Since $\dim K_3 = 2$, we have

   $$K_3 = \mathcal{N}(B-3I)^2 = \mathcal{N}\begin{pmatrix} 4 & 1 & -4 \\ 0 & 0 & 0 \\ 8 & 1 & -8 \end{pmatrix}^2 = \mathcal{N}\begin{pmatrix} -16 & 0 & 16 \\ 0 & 0 & 0 \\ -32 & 0 & 32 \end{pmatrix} = \operatorname{Span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

   which is spanned by a cycle of length two: $\left(\begin{smallmatrix} 1 \\ 0 \\ 1 \end{smallmatrix}\right) = (B-3I)\left(\begin{smallmatrix} 0 \\ 1 \\ 0 \end{smallmatrix}\right)$ is an eigenvector.

   Putting it together, $\beta = \left\{ \left(\begin{smallmatrix} 1 \\ 0 \\ 2 \end{smallmatrix}\right), \left(\begin{smallmatrix} 1 \\ 0 \\ 1 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 \\ 1 \\ 0 \end{smallmatrix}\right) \right\}$ is a Jordan canonical basis for B, and

   $$B = QJQ^{-1} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & 0 \end{pmatrix}^{-1}$$

3. Let $\mathrm{T} = \frac{\mathrm{d}}{\mathrm{d}x} \in \mathcal{L}(P_3(\mathbb{R}))$. With respect to the standard basis $\epsilon = \{1, x, x^2, x^3\}$, $[\mathrm{T}]_\epsilon = \left(\begin{smallmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{smallmatrix}\right)$.

   We have a single generalized eigenspace $K_0 = P_3(\mathbb{R})$. It is easy to check that $f(x) = x^3$ generates a cycle of length three and thus a Jordan canonical basis:

   $$\beta = \{6, 6x, 3x^2, x^3\} \implies [\mathrm{T}]_\epsilon = \begin{pmatrix} 6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1}$$

Our final results state that this process works in general.

Theorem 3.12. *Let* $\mathrm{T} \in \mathcal{L}(V)$ *have an eigenvalue* λ*. If* $\dim K_\lambda < \infty$*, then there exists a basis* $\beta_\lambda = \beta_{\mathbf{x}_1} \cup \cdots \cup \beta_{\mathbf{x}_n}$ *of* K_λ *consisting of finitely many linearly independent cycles.*

Intuition suggests that we start with a basis of the eigenspace E_λ and extend backwards: for each $\mathbf{x}$, if $\mathbf{x} = (\mathrm{T} - \lambda\mathrm{I})(\mathbf{y})$, then $\mathbf{x} \in \beta_{\mathbf{y}}$; now repeat until you have a maximum length cycle. This is essentially what we do, though a sneaky induction is required to make sure we keep track of everything and guarantee that the result really is a basis of K_λ.

Proof. We prove by induction on $m = \dim K_\lambda$.

Base case ($m = 1$) $K_\lambda = E_\lambda = \operatorname{Span} \mathbf{x}$ for some eigenvector $\mathbf{x}$. Plainly $\{\mathbf{x}\} = \beta_{\mathbf{x}}$ does the job.

Induction step Fix $m \geq 2$, and suppose *every* generalized eigenspace with dimension $< m$ (for *any* linear map!) has a basis consisting of independent cycles of generalized eigenvectors.

Write $n = \dim E_\lambda \leq m$ and $\mathrm{U} = (\mathrm{T} - \lambda\mathrm{I})_{K_\lambda}$. We break the argument into several steps.

1. Define $W = E_\lambda \cap \mathcal{R}(\mathrm{U})$: that is

$$\mathbf{w} \in W \iff \mathrm{U}(\mathbf{w}) = \mathbf{0} \text{ and } \mathbf{w} = \mathrm{U}(\mathbf{v}) \text{ for some } \mathbf{v} \in K_\lambda$$

Let $k = \dim W$, choose a subspace X such that $E_\lambda = W \oplus X$ and select a basis $\{\mathbf{x}_{k+1}, \ldots, \mathbf{x}_n\}$ of X. If $k = 0$, the induction step is finished (Exercise 8 shows that $K_\lambda = E_\lambda \ldots$).

2. Otherwise, take $j = 1$ in the proof of Lemma 3.10 to see that $\mathcal{R}(\mathrm{U})$ is T-invariant; it follows that $\mathcal{R}(U) = \widetilde{K}_\lambda$ is the single generalized eigenspace of $\mathrm{T}_{\mathcal{R}(\mathrm{U})}$. By the rank–nullity theorem,

$$\dim \widetilde{K}_\lambda = \dim \mathcal{R}(\mathrm{U}) = \operatorname{rank} \mathrm{U} = \dim K_\lambda - \operatorname{null} \mathrm{U} = m - \dim E_\lambda < m$$

By the induction hypothesis, $\mathcal{R}(\mathrm{U})$ has a basis of independent cycles. Since the last non-zero element in each cycle is an eigenvector, this basis consists of k distinct cycles $\beta_{\hat{\mathbf{x}}_1} \cup \cdots \cup \beta_{\hat{\mathbf{x}}_k}$ whose terminal vectors form a basis of W.

3. Since each $\hat{\mathbf{x}}_j \in \mathcal{R}(\mathrm{U})$, there exist vectors $\mathbf{x}_1, \ldots, \mathbf{x}_k$ such that $\hat{\mathbf{x}}_j = \mathrm{U}(\mathbf{x}_j)$. Including the length-one cycles generated by the basis of X, the cycles $\beta_{\mathbf{x}_1}, \ldots, \beta_{\mathbf{x}_n}$ now contain

$$\dim \mathcal{R}(\mathrm{U}) + k + (n - k) = \operatorname{rank} \mathrm{U} + \operatorname{null} \mathrm{U} = m$$

vectors. Exercise 8 verifies that these vectors are linearly independent. ∎

Recall (Theorem 3.5) that if the characteristic polynomial of $\mathrm{T} \in \mathcal{L}(V)$ splits, then $V = K_{\lambda_1} \oplus \cdots \oplus K_{\lambda_k}$ is the direct sum of generalized eigenspaces. But each of these now has a basis β_λ consisting of finitely many cycles. By Lemma 3.10, the matrix $[\mathrm{T}_{K_\lambda}]_{\beta_\lambda}$ is a Jordan canonical form, whence $\beta = \bigcup \beta_\lambda$ is a Jordan canonical basis for T. In conclusion:

Corollary 3.13. *If the characteristic polynomial of* T *splits, then it has a Jordan canonical basis.*

Exercises 3.1. *Key concepts: Jordan blocks, canonical forms & bases, Generalized eigenvectors & spaces, Building Jordan bases using cycles of generalized eigenvectors*

1. Compute the generalized eigenspaces K_λ, find bases consisting of unions of disjoint cycles of generalized eigenvectors, and thus find a Jordan canonical form J and invertible Q so as to express the matrix in the form QJQ^{-1}.

   (a) $A = \begin{pmatrix} 1 & 1 \\ -1 & 3 \end{pmatrix}$ (b) $B = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$ (c) $C = \begin{pmatrix} 11 & -4 & -5 \\ 21 & -8 & -11 \\ 3 & -1 & 0 \end{pmatrix}$ (d) $D = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 1 & -1 & 3 \end{pmatrix}$

2. If $\beta = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a Jordan canonical basis, what can you say about $\mathbf{v}_1$?

3. Briefly explain why matrix $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has no Jordan canonical form.

4. Find a Jordan canonical basis for each linear map T:

   (a) $\mathrm{T} \in \mathcal{L}(P_2(\mathbb{R}))$ defined by $\mathrm{T}(f(x)) = 2f(x) - f'(x)$

   (b) $\mathrm{T}(f) = f'$ defined on $\operatorname{Span}\{1, t, t^2, e^t, te^t\}$

   (c) $\mathrm{T}(A) = 2A + A^T$ defined on $M_2(\mathbb{R})$

5. In Example 3.11.1, suppose $\mathbf{x} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$. Show that almost any choice of a, b, c produces a Jordan canonical basis $\beta_{\mathbf{x}}$.

6. Complete the proof of Lemma 3.7.

   (a) Prove part 1, that $E_\lambda \le K_\lambda \le V$.

   (b) Verify that $\mathrm{T} - \mu\mathrm{I}$ and $\mathrm{T} - \lambda\mathrm{I}$ commute.

   (c) Prove part 3(b): generalized eigenspaces for distinct eigenvalues have trivial intersection.

7. Consider the induction step in the proof of Theorem 3.5.

   (a) Prove that W is T-invariant.

   (b) Explain why $W \cap K_\mu = \{\mathbf{0}\}$.

   (c) The assumption $p_W(t) = (\lambda_1 - t)^{\dim K_{\lambda_1}} \cdots (\lambda_k - t)^{\dim K_{\lambda_k}}$ near the end of the proof is the induction hypothesis for part 1(b). Why can't we also assume that $\dim K_{\lambda_j} = m_j$ and thus tidy the inequality argument near the end of the proof?

8. We finish some of the details of Theorem 3.12.

   (a) In step 1, suppose $\dim W = k = 0$. Explain why $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ is in fact a basis of K_λ, so that the rest of the proof is unnecessary.

   (b) In step 3, prove that the m vectors in the cycles $\beta_{\mathbf{x}_1}, \dots, \beta_{\mathbf{x}_n}$ are linearly independent. (*Hint: model your argument on part 1 of Lemma 3.10*)

3.2 Cycle Patterns and the Dot Diagram

In this section we develop a method for computing Jordan forms more efficiently and systematically. We start with a lengthy example that provides some clues how to proceed.

Example 3.14. Precisely three Jordan canonical forms $A, B, C \in M_3(\mathbb{R})$ correspond to the characteristic polynomial $p(t) = (5-t)^3$:

$$A = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} \qquad B = \begin{pmatrix} 5 & 1 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} \qquad C = \begin{pmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{pmatrix}$$

In all three cases the standard basis $\beta = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is Jordan canonical, so how do we distinguish things? By considering the *number* and *lengths* of the cycles of generalized eigenvectors.

- A has eigenspace $E_5 = K_5 = \mathbb{R}^3$. Since $(A - 5I)\mathbf{v} = \mathbf{0}$ for all $\mathbf{v} \in \mathbb{R}^3$, we have maximum cycle-length one. We therefore need three distinct cycles to construct a Jordan basis, e.g.

$$\beta_{\mathbf{e}_1} = \{\mathbf{e}_1\}, \quad \beta_{\mathbf{e}_2} = \{\mathbf{e}_2\}, \quad \beta_{\mathbf{e}_3} = \{\mathbf{e}_3\} \implies \beta = \beta_{\mathbf{e}_1} \cup \beta_{\mathbf{e}_2} \cup \beta_{\mathbf{e}_3} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$$

- B has eigenspace $E_5 = \operatorname{Span}\{\mathbf{e}_1, \mathbf{e}_3\}$. Provided $\mathbf{v} \notin E_5$,

$$\mathbf{v} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \implies (B - 5I)\mathbf{v} = \begin{pmatrix} b \\ 0 \\ 0 \end{pmatrix} \implies (B - 5I)^2\mathbf{v} = \mathbf{0}$$

  shows that $\beta_{\mathbf{v}}$ is a cycle with maximum length two. We therefore need two distinct cycles with lengths two and one to construct a Jordan basis, e.g.

$$\beta_{\mathbf{e}_2} = \{(B - 5I)\mathbf{e}_2,\ \mathbf{e}_2\} = \{\mathbf{e}_1, \mathbf{e}_2\}, \quad \beta_{\mathbf{e}_3} = \{\mathbf{e}_3\} \implies \beta = \beta_{\mathbf{e}_2} \cup \beta_{\mathbf{e}_3} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$$

- C has eigenspace $E_5 = \operatorname{Span} \mathbf{e}_1$. This time

$$\mathbf{v} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \implies (C - 5I)\mathbf{v} = \begin{pmatrix} b \\ c \\ 0 \end{pmatrix}, \quad (C - 5I)^2\mathbf{v} = \begin{pmatrix} c \\ 0 \\ 0 \end{pmatrix}, \quad (C - 5I)^3\mathbf{v} = \mathbf{0}$$

  Provided $c \neq 0$, this generates a cycle with maximum length three. Indeed this cycle is a Jordan basis, so one cycle is all we need:

$$\beta = \beta_{\mathbf{e}_3} = \{(C - 5I)^2\mathbf{e}_3,\ (C - 5I)\mathbf{e}_3,\ \mathbf{e}_3\} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$$

Why is the example relevant? If $\mathrm{T} \in \mathcal{L}(V)$ has characteristic polynomial $p(t) = (5-t)^3$, Theorem 3.12 tells us that T has a Jordan canonical form: one of the matrices A, B, C. Our goal is to develop a method whereby the pattern of cycle-lengths can be determined, thus allowing us to discern which Jordan form is correct. As a side-effect, this will also demonstrate that the pattern of cycle lengths for a given T is independent of the Jordan basis so that (modulo some reasonable restriction) the Jordan *form* of T is unique. To aid us in this endeavor, we require some terminology…

Definition 3.15. Let K_λ be a generalized eigenspace of $\mathrm{T} \in \mathcal{L}(V)$ and suppose $\beta_\lambda = \beta_{\mathbf{x}_1} \cup \cdots \cup \beta_{\mathbf{x}_n}$ is a Jordan canonical basis of T_{K_λ}, whose cycles are arranged in *non-increasing* length. That is:

1. $\beta_{\mathbf{x}_j} = \{(\mathrm{T} - \lambda \mathrm{I})^{k_j - 1}(\mathbf{x}_j), \ldots, \mathbf{x}_j\}$ has length k_j, and,
2. $k_1 \geq k_2 \geq \cdots \geq k_n$

The *dot diagram* of T_{K_λ} is a representation of the elements of β_λ, one dot for each vector: the j^{th} column represents the elements of $\beta_{\mathbf{x}_j}$ arranged vertically with $\mathbf{x}_j$ at the bottom.

Given a linear map, our goal is to identify the dot diagram as an intermediate step in the computation of a Jordan basis. First we observe how the conversion of dot diagrams to Jordan forms is essentially trivial.

Example 3.16. Suppose $\mathrm{T} \in \mathcal{L}(\mathbb{R}^{12})$ has the following eigenvalues and dot diagrams:

$$\begin{array}{ccc} \lambda_1 = -4 & \lambda_2 = 7 & \lambda_3 = 9 \\ \begin{array}{cc} \bullet & \bullet \\ \bullet & \bullet \\ \bullet & \end{array} & \begin{array}{cc} \bullet & \bullet \\ \bullet & \\ \bullet & \end{array} & \begin{array}{ccc} \bullet & \bullet & \bullet \\ & & \\ & & \end{array} \end{array}$$

Consider the diagram for $\lambda_1 = -4$. Each dot depicts a single vector in basis $\beta_{-4} = \beta_{\mathbf{x}_1} \cup \beta_{\mathbf{x}_2}$ of K_{-4}, as shown on the right.

$$\begin{array}{cc} (\mathrm{T}+4\mathrm{I})^2\mathbf{x}_1 & (\mathrm{T}+4\mathrm{I})\mathbf{x}_2 \\ (\mathrm{T}+4\mathrm{I})\mathbf{x}_1 & \mathbf{x}_2 \\ \mathbf{x}_1 & \end{array}$$

(a) The columns are the cycles: $\dim K_{-4} = 3 + 2 = 5$.

(b) The top row is the eigenspace: $E_{-4} = \operatorname{Span}\{(\mathrm{T}+4\mathrm{I})^2\mathbf{x}_1, (\mathrm{T}+4\mathrm{I})\mathbf{x}_2\}$ has dimension 2.

(c) The maximum cycle length is three: $K_{-4} = \mathcal{N}(\mathrm{T}+4\mathrm{I})^3$.

Similarly for $\lambda_2 = 7$: $\dim K_7 = 4$, $\dim E_7 = 2$ and $K_7 = \mathcal{N}(\mathrm{T} - 7\mathrm{I})^3$.
Finally, for $\lambda_3 = 9$ the generalized eigenspace is the eigenspace: $K_9 = E_9 = \mathcal{N}(\mathrm{T} - 9\mathrm{I})$ has dimension 3.
The map T has a Jordan canonical basis β with respect to which its Jordan canonical form is

$$[\mathrm{T}]_\beta = \begin{pmatrix} -4 & 1 & 0 & & & & & & & & & \\ 0 & -4 & 1 & & & & & & & & & \\ 0 & 0 & -4 & & & & & & & & & \\ & & & -4 & 1 & & & & & & & \\ & & & 0 & -4 & & & & & & & \\ & & & & & 7 & 1 & 0 & & & & \\ & & & & & 0 & 7 & 1 & & & & \\ & & & & & 0 & 0 & 7 & & & & \\ & & & & & & & & 7 & & & \\ & & & & & & & & & 9 & & \\ & & & & & & & & & & 9 & \\ & & & & & & & & & & & 9 \end{pmatrix}$$

Note how the sizes of the Jordan blocks are *non-increasing* within each eigenvalue.

Theorem 3.17. *Suppose β_λ is a Jordan canonical basis of T_{K_λ} as described in Definition 3.15, and suppose the i^{th} row of the dot diagram has r_i entries. Then:*

1. *For each $r \in \mathbb{N}$, the vectors associated to the dots in the first r rows form a basis of $\mathcal{N}(\mathrm{T}-\lambda\mathrm{I})^r$.*
2. *The dimension of the eigenspace is $r_1 = \mathrm{null}(\mathrm{T}-\lambda\mathrm{I}) = \dim V - \mathrm{rank}(\mathrm{T}-\lambda\mathrm{I})$.*
3. *When $i > 1$, $r_i = \mathrm{null}(\mathrm{T}-\lambda\mathrm{I})^i - \mathrm{null}(\mathrm{T}-\lambda\mathrm{I})^{i-1} = \mathrm{rank}(\mathrm{T}-\lambda\mathrm{I})^{i-1} - \mathrm{rank}(\mathrm{T}-\lambda\mathrm{I})^i$*

Example (3.14 cont). We describe the dot diagrams of the three matrices A, B, C, the corresponding vectors in the Jordan canonical basis β, and the values r_i.

A : • • • $\quad \mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3$

Since $A - 5I$ is the zero matrix, $r_1 = 3 - \mathrm{rank}(A - 5I) = 3$. The dot diagram has one row, corresponding to three independent cycles of length one: $\beta = \beta_{\mathbf{e}_1} \cup \beta_{\mathbf{e}_2} \cup \beta_{\mathbf{e}_3}$.

B : • • $\quad \mathbf{e}_1 \quad \mathbf{e}_3 \qquad (B-5I)\mathbf{e}_2 \quad \mathbf{e}_3$
• $\quad \mathbf{e}_2 \qquad \mathbf{e}_2$

Row 1: $B-5I = \begin{pmatrix} 0&1&0\\0&0&0\\0&0&0 \end{pmatrix}$ has rank 1, whence $r_1 = 3-1 = 2$. This row $\{\mathbf{e}_1, \mathbf{e}_3\}$ spans $E_5 = \mathcal{N}(B-5I)$.

Row 2: $(B-5I)^2$ is the zero matrix, whence $r_2 = \mathrm{rank}(B-5I) - \mathrm{rank}(B-5I)^2 = 1 - 0 = 1$.

The dot diagram corresponds to $\beta = \beta_{\mathbf{e}_2} \cup \beta_{\mathbf{e}_3} = \{\mathbf{e}_1, \mathbf{e}_2\} \cup \{\mathbf{e}_3\}$.

C : • $\quad \mathbf{e}_1 \qquad (C-5I)^2\mathbf{e}_3$
• $\quad \mathbf{e}_2 \qquad (C-5I)\mathbf{e}_3$
• $\quad \mathbf{e}_3 \qquad \mathbf{e}_3$

Row 1: $C-5I = \begin{pmatrix} 0&1&0\\0&0&1\\0&0&0 \end{pmatrix}$ has rank 2, whence $r_1 = 3-2 = 1$. This row $\{\mathbf{e}_1\}$ spans $E_5 = \mathcal{N}(C-5I)$.

Row 2: $(C-5I)^2 = \begin{pmatrix} 0&0&1\\0&0&0\\0&0&0 \end{pmatrix}$ has rank 1, whence $r_2 = 2-1 = 1$. The first two rows $\{\mathbf{e}_1, \mathbf{e}_2\}$ form a basis of $\mathcal{N}(C-5I)^2$.

Row 3: $(C-5I)^3$ is the zero matrix, whence $r_3 = \mathrm{rank}(C-5I)^2 - \mathrm{rank}(C-5I)^3 = 1 - 0 = 1$.

Proof. As previously, let $\mathrm{U} = \mathrm{T} - \lambda\mathrm{I}$.

1. Since each dot represents a basis vector $\mathrm{U}^p(\mathbf{v}_j)$, any $\mathbf{v} \in K_\lambda$ may be written uniquely as a linear combination of the dots. Applying U moves all dots up a row and all dots in the top row to $\mathbf{0}$. It follows that $\mathbf{v} \in \mathcal{N}(\mathrm{U}^r) \iff$ it lies in the span of the first r rows. Since the dots are linearly independent, they form a basis.
2. By part 1, $r_1 = \dim \mathcal{N}(\mathrm{U}) = \mathrm{null}(\mathrm{T}-\lambda\mathrm{I}) = \dim V - \mathrm{rank}(\mathrm{T}-\lambda\mathrm{I})$.
3. More generally,

$$\begin{aligned} r_i &= (r_1 + \cdots + r_i) - (r_1 + \cdots + r_{i-1}) = \dim \mathcal{N}(\mathrm{U}^i) - \dim \mathcal{N}(\mathrm{U}^{i-1}) \\ &= \mathrm{null}(\mathrm{U}^i) - \mathrm{null}(\mathrm{U}^{i-1}) = \mathrm{rank}(\mathrm{T}-\lambda\mathrm{I})^{i-1} - \mathrm{rank}(\mathrm{T}-\lambda\mathrm{I})^i \end{aligned}$$

■

Since the ranks of the maps $(\mathrm{T}-\lambda\mathrm{I})^i$ are independent of basis, so also is the dot diagram…

Corollary 3.18. *For any eigenvalue λ, the dot diagram is uniquely determined by* T *and λ. If Jordan blocks for each eigenspace are listed in non-increasing order, then the Jordan form of a linear map is unique up to the order of its eigenvalues.*

We now have a slightly more systematic method for finding Jordan canonical bases.

Example 3.19. The matrix $A = \left(\begin{smallmatrix} 6 & 2 & -4 & -6 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 2 & 1 & -2 & -1 \end{smallmatrix}\right)$ has characteristic polynomial $p(t) = (2-t)^1(3-t)^3$, and two generalized eigenspaces:

$(\lambda = 2)$ $K_2 = \mathcal{N}(A-2I)^1$ is the one-dimensional eigenspace itself! The trivial dot diagram • corresponds to a single eigenvector in

$$E_2 = \mathcal{N}\begin{pmatrix} 4 & 2 & -4 & -6 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 2 & 1 & -2 & -3 \end{pmatrix} = \operatorname{Span}\begin{pmatrix} 3 \\ 0 \\ 0 \\ 2 \end{pmatrix}$$

$(\lambda = 3)$ $K_3 = \mathcal{N}(A-3I)^3$. To find the dot diagram, compute powers of $A-3I$:

Row 1: $A - 3I = \left(\begin{smallmatrix} 3 & 2 & -4 & -6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 1 & -2 & -4 \end{smallmatrix}\right)$ has rank 2 and the first row has $r_1 = 4-2 = 2$ entries.

Row 2: $(A-3I)^2 = \left(\begin{smallmatrix} -3 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -2 & 0 & 0 & 4 \end{smallmatrix}\right)$ has rank 1 and the second row has $r_2 = 2-1 = 1$ entry.

Since we now have three dots (= $\dim K_3$), the algorithm terminates and we conclude that the dot diagram for K_3 is

$$\begin{matrix} \bullet & \bullet \\ \bullet & \end{matrix}$$

For the single dot in the second row, choose anything in $\mathcal{N}(A-3I)^2$ that isn't an eigenvector; perhaps the simplest choice is $\mathbf{x}_1 = \mathbf{e}_2$, which yields the two-cycle

$$\beta_{\mathbf{x}_1} = \{(A-3I)\mathbf{x}_1, \mathbf{x}_1\} = \left\{\left(\begin{smallmatrix} 2 \\ 0 \\ 0 \\ 1 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 \\ 1 \\ 0 \\ 0 \end{smallmatrix}\right)\right\}$$

To finish the first row, choose any eigenvector to complete the span: for instance $\mathbf{x}_2 = \left(\begin{smallmatrix} 0 \\ 2 \\ 1 \\ 0 \end{smallmatrix}\right)$.

We now have suitable cycles and a Jordan canonical basis/form:

$$\beta = \left\{\begin{pmatrix} 3 \\ 0 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 1 \\ 0 \end{pmatrix}\right\}, \qquad A = QJQ^{-1} = \begin{pmatrix} 3 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \\ 2 & 1 & 0 & 0 \end{pmatrix}\begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}\begin{pmatrix} 3 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \\ 2 & 1 & 0 & 0 \end{pmatrix}^{-1}$$

Other choices of basis are available: for instance, we could generate a two-cycle $\beta_{\mathbf{x}_1}$ using any vector $\mathbf{x}_1 = a\mathbf{e}_2 + b\mathbf{e}_3 + c(2\mathbf{e}_1 + \mathbf{e}_4) \in \mathcal{N}(A-2I)^2$ for which a, b are not both zero…

Example 3.20. For variety, here is an example with a non-matrix map.

Let $\epsilon = \{1, x, y, x^2, y^2, xy\}$ and define the partial differential operator $\mathrm{T}\big(f(x,y)\big) = 2\frac{\partial f}{\partial x} - \frac{\partial f}{\partial y}$ on the real vector space $V = \operatorname{Span}_{\mathbb{R}} \epsilon$. We quickly compute:

$$[\mathrm{T}]_\epsilon = \begin{pmatrix} 0 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & -1 \\ 0 & 0 & 0 & 0 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \implies p(t) = t^6, \quad [\mathrm{T}^2]_\epsilon = \begin{pmatrix} 0 & 0 & 0 & 8 & 2 & -4 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad [\mathrm{T}^3]_\epsilon = O$$

There is only one eigenvalue $\lambda = 0$ and therefore one generalized eigenspace $K_0 = V$. We could keep working with matrices, but it is easy to translate their nullspaces back to subspaces of V, from which the necessary data can be read off:

$$\begin{aligned} &\mathcal{N}(\mathrm{T}) = \operatorname{Span}\{1, x+2y, x^2+4y^2+4xy\} && \operatorname{null}\mathrm{T} = 3,\ \operatorname{rank}\mathrm{T} = 3,\ r_1 = 3 \\ &\mathcal{N}(\mathrm{T}^2) = \operatorname{Span}\{1, x, y, x^2+2xy, 2y^2+xy\} && \operatorname{null}\mathrm{T}^2 = 5,\ \operatorname{rank}\mathrm{T}^2 = 1,\ r_2 = 3-1 = 2 \end{aligned}$$

These calculations produce five dots; since $\dim K_0 = 6$, the last row has one dot and the dot diagram is as shown.

$$\begin{array}{ccc} \bullet & \bullet & \bullet \\ \bullet & \bullet & \\ \bullet & & \end{array}$$

Since the first two rows span $\mathcal{N}(\mathrm{T}^2)$, we may choose any $f_1 \notin \mathcal{N}(\mathrm{T}^2)$ for the final dot. A suitable choice is $f_1 = xy$, from which the first column of the dot diagram becomes

$$\begin{array}{ccc} \mathrm{T}^2(xy) & \bullet & \bullet \\ \mathrm{T}(xy) & \bullet & \\ xy & & \end{array} \qquad \begin{array}{ccc} -4 & \bullet & \bullet \\ 2y-x & \bullet & \\ xy & & \end{array}$$

Now choose the second dot on the second row to be anything in $\mathcal{N}(\mathrm{T}^2)$ such that the first two rows span $\mathcal{N}(\mathrm{T}^2)$. This time $f_2 = x^2 - 4y^2$ is suitable and the diagram becomes

$$\begin{array}{ccc} \mathrm{T}^2(xy) & \mathrm{T}(x^2-4y^2) & \bullet \\ \mathrm{T}(xy) & x^2-4y^2 & \\ xy & & \end{array} \qquad \begin{array}{ccc} -4 & 4x+8y & \bullet \\ 2y-x & x^2-4y^2 & \\ xy & & \end{array}$$

The final dot is chosen so that the first row spans $\mathcal{N}(\mathrm{T})$: this time $f_3 = x^2 + 4y^2 + 4xy$ works. The result is a Jordan canonical basis and form for T:

$$\beta = \{-4,\ 2y-x,\ xy,\ 4x+8y,\ x^2-4y^2,\ x^2+4y^2+4xy\}, \qquad J = [\mathrm{T}]_\beta = \left(\begin{array}{ccc|cc|c} 0 & 1 & 0 & & & \\ 0 & 0 & 1 & & & \\ 0 & 0 & 0 & & & \\ \hline & & & 0 & 1 & \\ & & & 0 & 0 & \\ \hline & & & & & 0 \end{array}\right)$$

As previously, other choices of cycle-generators f_1, f_2, f_3 are available. While alternatives result in different Jordan bases, Corollary 3.18 assures us that we'll always obtain the same canonical form J.

Exercises 3.2. *Key concepts: Dot diagrams: meaning & computation, Uniqueness of Jordan form*

1. Let T be a linear operator whose characteristic polynomial splits. Suppose the eigenvalues and the dot diagrams for the generalized eigenspaces K_{λ_i} are as follows:

$$\begin{array}{ccc} \lambda_1 = 2 & \lambda_2 = 4 & \lambda_3 = -3 \\ \bullet\ \bullet\ \bullet & \bullet\ \bullet & \bullet\ \bullet \\ \bullet\ \bullet\ & \bullet\ & \\ \bullet\ \ & \bullet\ & \end{array}$$

   Describe the Jordan form J of T.

2. Suppose T has Jordan canonical form J.

$$J = \begin{pmatrix} 2 & 1 & 0 & & & & \\ 0 & 2 & 1 & & & & \\ 0 & 0 & 2 & & & & \\ & & & 2 & 1 & & \\ & & & 0 & 2 & & \\ & & & & & 3 & \\ & & & & & & 3 \end{pmatrix}$$

   (a) State the characteristic polynomial of T.

   (b) For each eigenvalue λ, find the dot diagram and the smallest k such that $K_\lambda = \mathcal{N}(\mathrm{T} - \lambda \mathrm{I})^k$.

3. In Example 3.19, compute the two-cycle $\beta_{\mathbf{x}_1}$ generated by $\mathbf{x}_1 = \mathbf{e}_3$. Hence obtain an alternative Jordan canonical basis.

4. For each matrix A find a Jordan canonical form and an invertible Q such that $A = QJQ^{-1}$.

   (a) $A = \begin{pmatrix} -3 & 3 & -2 \\ -7 & 6 & -3 \\ 1 & -1 & 2 \end{pmatrix}$ (b) $A = \begin{pmatrix} 0 & 1 & -1 \\ -4 & 4 & -2 \\ -2 & 1 & 1 \end{pmatrix}$ (c) $A = \begin{pmatrix} 0 & -3 & 1 & 2 \\ -2 & 1 & -1 & 2 \\ -2 & 1 & -1 & 2 \\ -2 & -3 & 1 & 4 \end{pmatrix}$

5. For each linear operator T, find a Jordan canonical form J and basis β:

   (a) $\mathrm{T}(f) = f'$ on $\operatorname{Span}_{\mathbb{R}} \{e^t, te^t, t^2e^t, e^{2t}\}$

   (b) $\mathrm{T}\big(f(x)\big) = xf''(x)$ on $P_3(\mathbb{R})$

   (c) $\mathrm{T}(f) = af_x + bf_y$ on $\operatorname{Span}_{\mathbb{R}} \{1, x, y, x^2, y^2, xy\}$. How does your answer depend on a, b?

6. (Generalized Eigenvector Method for ODEs) Let $A \in M_n(\mathbb{R})$ have an eigenvalue λ and suppose $\beta_{\mathbf{v}_0} = \{\mathbf{v}_{k-1}, \ldots, \mathbf{v}_1, \mathbf{v}_0\}$ is a cycle of generalized eigenvectors for this eigenvalue.

   (a) Show that $\mathbf{x}(t) := e^{\lambda t} \sum_{j=0}^{k-1} b_j(t)\mathbf{v}_j$ is a solution to $\mathbf{x}'(t) = A\mathbf{x}$ if and only if

$$\begin{cases} b_0'(t) = 0, \text{ and} \\ b_j'(t) = b_{j-1}(t) \text{ when } j \geq 1 \end{cases}$$

   (b) Use this method to find the general solution to the system of differential equations

$$\mathbf{x}' = \begin{pmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix} \mathbf{x}$$